

The Chain Rule

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Fall 2026

Recall that in calculus I, we used the chain rule to find the derivatives of composite functions.

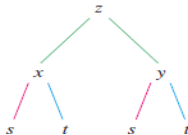
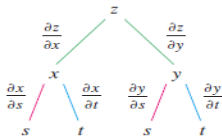
The Chain Rule: Case I Suppose that $z = f(x, y)$ is a differentiable function of x and y , where $x = g(t)$ and $y = h(t)$ are both differentiable functions of t . Then z is a differentiable function of t . Moreover

$$\frac{dz}{dt} = \frac{\partial z}{\partial x} \frac{dx}{dt} + \frac{\partial z}{\partial y} \frac{dy}{dt}$$

The Chain Rule: Case II Suppose that $z = f(x, y)$, $x = g(s, t)$ and $y = h(s, t)$ where all functions f, g, h are differentiable. Then z is a differentiable function of s and t . Moreover

$$\frac{\partial z}{\partial s} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial s} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial s} \quad \text{and} \quad \frac{\partial z}{\partial t} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial t} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial t}$$

For the chain rule, we can make use of the **tree notation**.



Example I. Use the chain rule to find $\frac{dz}{dt}$ or $\frac{dw}{dt}$ if

I. $z = \sqrt{x^2 + y^2}$, $x = e^{2t}$, $y = e^{-2t}$

II. $w = xy + yz^2$, $x = e^t$, $y = e^t \sin(t)$, $z = e^t \cos(t)$

III. $z = \sec(x + y) \sin(x - y^2)$, $x = \pi t$, $y = \sqrt{t}$

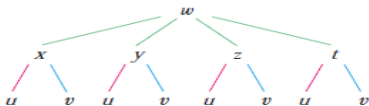
Solution

$$\begin{aligned} 1. \quad \frac{dz}{dt} &= \frac{\partial z}{\partial x} \frac{dx}{dt} + \frac{\partial z}{\partial y} \frac{dy}{dt} = \frac{1}{2} (x^2 + y^2)^{-1/2} 2x e^{2t} \cdot 2 + \frac{1}{2} (x^2 + y^2)^{-1/2} 2y e^{-2t} (-2) \\ &= \frac{2xe^{2t} - 2ye^{2t}}{\sqrt{x^2 + y^2}} \end{aligned}$$

$$\begin{aligned} 2. \quad \frac{dw}{dt} &= \frac{\partial w}{\partial x} \frac{dx}{dt} + \frac{\partial w}{\partial y} \frac{dy}{dt} + \frac{\partial w}{\partial z} \frac{dz}{dt} \\ &= y e^t + (x + z^2) (e^t \cos t + e^t \sin t) + 2yz (-e^t \sin t + e^t \cos t) \end{aligned}$$

Note that the chain rule can be generalized to functions with more variables.

- $\models w = f(x, y, z, t), \quad x = x(u, v), \quad y = y(u, v), \quad t = t(u, v)$



- $\models u = u(x, y, z), \quad x = x(r, s, t), \quad y = y(r, s, t), \quad z = z(r, s, t)$



Example 2 Use the chain rule to find $\frac{\partial z}{\partial s}$ and $\frac{\partial z}{\partial t}$ if

I. $z = \frac{x}{y}$, $x = se^t$, $y = 1 + se^{-t}$

II. $z = e^{xy} \tan(y)$, $x = s + 2t$, $y = \frac{s}{t}$

Solution

$$1. \frac{\partial z}{\partial s} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial s} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial s} = \frac{1}{y} e^t + \left(-\frac{x}{y^2}\right) (e^{-t}) = \frac{1}{y} e^t - \frac{x}{y^2} e^{-t}$$

$$\frac{\partial z}{\partial t} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial t} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial t} = \frac{1}{y} (se^t) + \left(-\frac{x}{y^2}\right) (-se^{-t}) = \frac{s}{y} e^t + \frac{xs}{y^2} e^{-t}$$

$$2. \frac{\partial z}{\partial s} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial s} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial s} = ye^{xy} \tan y \cdot 1 + (e^{xy} \sec y + xe^{xy} \tan y) \cdot \frac{1}{t}$$

$$\frac{\partial z}{\partial t} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial t} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial t} = ye^{xy} \tan y \cdot 2 + (e^{xy} \sec y + xe^{xy} \tan y) \left(-\frac{s}{t^2}\right)$$

Example 3. If $z = f(x, y)$ is differentiable with $x = g(t)$, $y = h(t)$ with $g(3) = 2$, $g'(3) = 5$, $h(3) = 7$, $h'(3) = -4$, $f_x(2, 7) = 6$, $f_y(2, 7) = -8$.

Find $\frac{dz}{dt}$ when $t = 3$.

Solution

$$\begin{aligned}\frac{dz}{dt} &= \frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt} = f_x(2, 7)g'(3) + f_y(2, 7)h'(3) \\ &= 6 \cdot 5 + (-8)(-4) = 62\end{aligned}$$

Example 4. If $W = F(u(s, t), v(s, t))$ is differentiable with
 $u(1, 0) = 2$, $u_s(1, 0) = -2$, $u_t(1, 0) = 6$, $v(1, 0) = 3$, $v_s(1, 0) = 5$, $v_t(1, 0) = 4$, $F_u(2, 3) = -1$, $F_v(2, 3) = 10$, find $W_s(1, 0)$ and $W_t(1, 0)$.

Solution

$$\frac{\partial W}{\partial s} = \frac{\partial W}{\partial u} \frac{\partial u}{\partial s} + \frac{\partial W}{\partial v} \frac{\partial v}{\partial s}$$

$$W_s(1, 0) = F_u(2, 3) u_s(1, 0) + F_v(2, 3) v_s(1, 0)$$

$$= (-1)(-2) + 10 \cdot 5 = 52$$

Similarly, $W_t(1, 0) = 34$

Example 5. Assume that all the given functions below are differentiable. Use a tree diagram to write the chain rule for the given case.

- $z = f(x, y), \quad x = x(r, s, t), \quad y = y(r, s, t)$
- $w = f(x, y, z), \quad x = x(t, u), \quad y = y(t, u), \quad z = z(t, u)$

Example 6. Use the chain rule to find the given partial derivatives.

○ If $u = \sqrt{r^2 + s^2}$, $r = y + x \cos(t)$, $s = x + y \sin(t)$, find $\frac{\partial u}{\partial x}$, $\frac{\partial u}{\partial y}$, $\frac{\partial u}{\partial t}$ when $x = 1$, $y = 2$, $t = 0$.

○ If $M = xe^{y-z^2}$, $x = 2uv$, $y = u - v$, $z = u + v$, find $\frac{\partial M}{\partial u}$, $\frac{\partial M}{\partial v}$ when $u = 3$, $v = -1$.

Solution

$$\frac{\partial u}{\partial x} = \frac{\partial u}{\partial r} \frac{\partial r}{\partial x} + \frac{\partial u}{\partial s} \frac{\partial s}{\partial x} = \frac{1}{2} (r^2 + s^2)^{-1/2} \cdot 2r \cos t + \frac{1}{2} (r^2 + s^2)^{-1/2} \cdot 2s \cdot 1$$

when $x=1, y=2, t=0 \Rightarrow r=3, s=1$ so $\frac{\partial u}{\partial x} = \frac{4}{\sqrt{10}}$

$$\frac{\partial u}{\partial t} = \frac{\partial u}{\partial r} \frac{\partial r}{\partial t} + \frac{\partial u}{\partial s} \frac{\partial s}{\partial t} = \frac{1}{2} (r^2 + s^2)^{-1/2} \cdot 2r(-x \sin t) + \frac{1}{2} (r^2 + s^2)^{-1/2} \cdot 2s \cdot y \cos t$$

when $x=1, y=2, t=0$,

$$\frac{\partial u}{\partial t} = \frac{2}{\sqrt{10}}$$

Implicit Differentiation

Proposition Consider an implicit function $F(x, y) = 0$.
Then we have

$$\frac{dy}{dx} = -\frac{\frac{\partial F}{\partial x}}{\frac{\partial F}{\partial y}} = -\frac{F_x}{F_y}$$

Proof.

Example 1. Find $\frac{dy}{dx}$ if

- I. $y^5 + x^2y^3 = 1 + ye^{x^2}$
- II. $\sin(x) + \cos(y) = \sin(x) \cos(y)$
- III. $\cos(x - y) = xe^y$

Solution

- I. Let $F(x, y) := y^5 + x^2y^3 - 1 - ye^{x^2} = 0$. Thus, we have

$$\frac{dy}{dx} = -\frac{F_x}{F_y} = -\frac{(2xy^3 - 2xye^{x^2})}{5y^4 + 3x^2y^2 - e^{x^2}}$$

- II. Let $F(x, y) := \sin(x) + \cos(y) - \sin(x) \cos(y) = 0$. Thus, we have

$$\frac{dy}{dx} = -\frac{F_x}{F_y} = -\frac{(\cos(x) - \cos(x) \cos(y))}{-\sin(y) + \sin(x) \sin(y)}.$$

Implicit Differentiation

Proposition Consider an implicit function $F(x, y, z) = 0$.
Then we have

$$\frac{\partial z}{\partial x} = -\frac{\frac{\partial F}{\partial x}}{\frac{\partial F}{\partial z}} \quad \text{and} \quad \frac{\partial z}{\partial y} = -\frac{\frac{\partial F}{\partial y}}{\frac{\partial F}{\partial z}}$$

Example 8. Find $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$ if

- I. $xyz = \cos(x + y + z)$
- II. $\sin(x) + \cos(z) = \sin(x) \cos(y)$
- III. $yz = \ln(x + z)$

Solution

I. Let $F(x, y) := xyz - \cos(x + y + z) = 0$. Thus, we have

$$\frac{\partial z}{\partial x} = -\frac{F_x}{F_z} = -\frac{(yz + \sin(x + y + z))}{xy + \sin(x + y + z)}$$

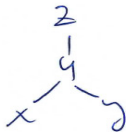
and

$$\frac{\partial z}{\partial y} = -\frac{F_y}{F_z} = -\frac{(xz + \sin(x + y + z))}{xy + \sin(x + y + z)}$$

Example 9. If $z = f(x - y)$, then show that $\frac{\partial z}{\partial x} + \frac{\partial z}{\partial y} = 0$.

Solution

$$\text{Let } u = x - y \Rightarrow$$



$$\frac{\partial z}{\partial x} = \frac{\partial z}{\partial u} \frac{\partial u}{\partial x} = \frac{\partial z}{\partial u}$$

$$\frac{\partial z}{\partial y} = \frac{\partial z}{\partial u} \frac{\partial u}{\partial y} = -\frac{\partial z}{\partial u}$$

$$\frac{\partial z}{\partial x} + \frac{\partial z}{\partial y} = 0$$

Further Problems

Partial derivatives are used mostly in the solutions of PDEs.

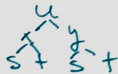
Example 10. If $u = f(x, y)$ where $x = e^s \cos(t)$ and $y = e^s \sin(t)$, show that

$$\left(\frac{\partial u}{\partial x}\right)^2 + \left(\frac{\partial u}{\partial y}\right)^2 = e^{-2s} \left[\left(\frac{\partial u}{\partial s}\right)^2 + \left(\frac{\partial u}{\partial t}\right)^2 \right]$$

$$u = f(x, y), \quad x = e^s \cos t, \quad y = e^s \sin t$$

$$u_s = u_x \cdot x_s + u_y \cdot y_s = u_x \cdot e^s \cos t + u_y \cdot e^s \sin t$$

$$u_t = u_x x_t + u_y y_t = u_x (-e^s \sin t) + u_y e^s \cos t$$



$$\begin{aligned} (u_s)^2 &= \left(e^s (\cos t u_x + \sin t u_y) \right)^2 = e^{2s} (\cos^2 t (u_x)^2 + \sin^2 t (u_y)^2 + 2 \sin t \cos t u_x u_y) \\ (u_t)^2 &= \left(e^s (-\sin t u_x + \cos t u_y) \right)^2 = e^{2s} (\sin^2 t (u_x)^2 + \cos^2 t (u_y)^2 - 2 \sin t \cos t u_x u_y) \end{aligned}$$

+

$$(u_s)^2 + (u_t)^2 = e^{2s} ((u_x)^2 + (u_y)^2) \Rightarrow$$

$$(u_x)^2 + (u_y)^2 = e^{-2s} ((u_s)^2 + (u_t)^2).$$

Example 11. Show that $z = f(x + at) + g(x - at)$ satisfies

$$\frac{\partial^2 z}{\partial t^2} = a^2 \frac{\partial^2 z}{\partial x^2}$$

where a is a constant.

Hint. Make the change of variables

$$u = x + at \quad \text{and} \quad v = x - at$$

Solution

$$z = f(x + at) + g(x - at) = f(u) + g(v), \quad \begin{matrix} u = x + at \\ v = x - at \end{matrix}$$



$$z_t = z_u \cdot u_t + z_v \cdot v_t = f'(u) \cdot a + g'(v) \cdot (-a) = af'(u) - ag'(v)$$

$$z_{tt} = (z_t)_t = af''(u) \cdot \overset{a}{u_t} - ag''(v) \cdot \overset{-a}{v_t} = a^2 f''(u) + a^2 g''(v)$$

$$z_x = z_u \cdot u_x + z_v \cdot v_x = f'(u) + g'(v)$$

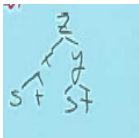
$$z_{xx} = (z_x)_x = f''(u) \cdot u_x + g''(v) \cdot v_x = f''(u) + g''(v)$$

$$\text{We have} \quad z_{tt} = a^2 (f''(u) + g''(v)) = a^2 z_{xx}.$$

Example 12. Let $z = f(x, y)$ where $x = s + t$ and $y = s - t$. Show that

$$\left(\frac{\partial z}{\partial x}\right)^2 - \left(\frac{\partial z}{\partial y}\right)^2 = \frac{\partial z}{\partial s} \frac{\partial z}{\partial t}$$

Solution



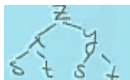
$$\frac{\partial z}{\partial s} = \frac{\partial z}{\partial x} + \frac{\partial z}{\partial y}, \quad \frac{\partial z}{\partial t} = \frac{\partial z}{\partial x} - \frac{\partial z}{\partial y}. \quad \text{Here}$$

$$\frac{\partial z}{\partial s} \cdot \frac{\partial z}{\partial t} = \left(\frac{\partial z}{\partial x} + \frac{\partial z}{\partial y} \right) \left(\frac{\partial z}{\partial x} - \frac{\partial z}{\partial y} \right) = \left(\frac{\partial z}{\partial x} \right)^2 - \left(\frac{\partial z}{\partial y} \right)^2$$

Example 13. Let $z = f(x, y)$ where $x = g(s, t)$ and $y = h(s, t)$. Find

$$\frac{\partial^2 z}{\partial t^2}, \quad \frac{\partial^2 z}{\partial s \partial t}, \quad \frac{\partial^2 z}{\partial s^2}$$

Solution



$$\frac{\partial z}{\partial t} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial t} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial t}$$

$$\frac{\partial^2 z}{\partial t^2} = \frac{\partial}{\partial t} \left(\frac{\partial z}{\partial x} \frac{\partial x}{\partial t} \right) + \frac{\partial}{\partial t} \left(\frac{\partial z}{\partial y} \frac{\partial y}{\partial t} \right)$$

$$= \frac{\partial}{\partial t} \left(\frac{\partial z}{\partial x} \right) \frac{\partial x}{\partial t} + \frac{\partial^2 z}{\partial x^2} \frac{\partial x}{\partial t} + \frac{\partial}{\partial t} \left(\frac{\partial z}{\partial y} \right) \frac{\partial y}{\partial t} + \frac{\partial^2 z}{\partial y^2} \frac{\partial y}{\partial t}$$

$$= \frac{\partial^2 z}{\partial x^2} \left(\frac{\partial x}{\partial t} \right)^2 + \frac{\partial^2 z}{\partial x \partial y} \frac{\partial x}{\partial t} \frac{\partial y}{\partial t} + \frac{\partial^2 z}{\partial y^2} \left(\frac{\partial y}{\partial t} \right)^2 + \frac{\partial^2 z}{\partial x \partial y} \frac{\partial y}{\partial t} \frac{\partial x}{\partial t} + \frac{\partial^2 z}{\partial y^2} \frac{\partial y}{\partial t}$$

$$= \frac{\partial^2 z}{\partial x^2} \left(\frac{\partial x}{\partial t} \right)^2 + 2 \frac{\partial^2 z}{\partial x \partial y} \frac{\partial x}{\partial t} \frac{\partial y}{\partial t} + \frac{\partial^2 z}{\partial y^2} \left(\frac{\partial y}{\partial t} \right)^2$$

$$+ \frac{\partial^2 x}{\partial t^2} \frac{\partial z}{\partial x} + \frac{\partial^2 y}{\partial t^2} \frac{\partial z}{\partial y}$$

Example 14. Suppose that f and g are two differentiable functions. If the function

$$F := xf(x+y) + yg(x+y)$$

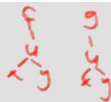
satisfies

$$\frac{\partial^2 F}{\partial x^2} + \frac{\partial^2 F}{\partial y^2} - \alpha^2 \frac{\partial^2 F}{\partial x \partial y} = 0$$

find value(s) of α .

$$F = xf(x+y) + yg(x+y). \text{ let } u := x+y \Rightarrow$$

$$F = xf(u) + yg(u).$$



$$F_x = f(u) + x f'(u) \cdot 1 + y g'(u) \cdot 1 \Rightarrow$$

$$F_y = x f'(u) + g(u) + y g'(u)$$

$$F_{xx} = (F_x)_x = f'(u) + f'(u) + x f''(u) + y g'(u) = 2f'(u) + x f''(u) + y g'(u)$$

$$F_{yy} = (F_y)_y = x f'(u) + g'(u) + g'(u) + y g''(u) = 2g'(u) + x f'(u) + y g''(u)$$

$$F_{yx} = (F_y)_x = f'(u) + x f''(u) + g'(u) + y g''(u)$$

$$F_{xx} + F_{yy} - \alpha^2 F_{yx} = 2f''(u) + x f''(u) + y g''(u) + 2g'(u) + x f'(u) + y g''(u) - \alpha^2 (f'(u) + g'(u) + x f''(u) + y g''(u)) = 0$$

$$f'(u)(2 - \alpha^2) + g'(u)(2 - \alpha^2) + f''(u)(2x - \alpha^2 x) + g''(u)(2y - \alpha^2 y) = 0$$

$$\Rightarrow \alpha^2 = 2 \Rightarrow \alpha = \pm \sqrt{2}$$

Example 15. Suppose that f is a twice differentiable function, and $\alpha \in \mathbb{R}$.

Let $u = f(x + \alpha t)$. If

$$\frac{\partial^2 u}{\partial t^2} - 9 \frac{\partial^2 u}{\partial x^2} = 0$$

, then find the value(s) of α .

Solution

$$u = f(x + \alpha t), \quad u_{tt} - 9u_{xx} = 0.$$

$$\text{Let } w := x + \alpha t \Rightarrow \begin{array}{c} u = f(w) \\ \quad \downarrow \\ x \quad t \end{array}$$

$$u_x = f'(w) \cdot 1 \Rightarrow u_{xx} = f''(w)$$

$$u_t = f'(w) \cdot \frac{\partial w}{\partial t} = \alpha f'(w), \quad u_{tt} = \alpha \left(f''(w) \cdot \frac{\partial w}{\partial t} \right) = \alpha^2 f''(w)$$

$$u_{tt} - 9u_{xx} = 0 \Rightarrow \alpha^2 f''(w) - 9 f''(w) = (\alpha^2 - 9) f''(w) = 0$$

$$\Rightarrow \alpha^2 = 9 \Rightarrow \alpha = \pm 3.$$

Example 16. If $f(x, y)$ is harmonic, show that $f(x^2 - y^2, 2xy)$ is also harmonic.

Solution. Let $u = x^2 - y^2$ and $v = 2xy$. If $z = f(u, v)$, then

$$\frac{\partial z}{\partial x} = 2xf_u(u, v) + 2yf_v(u, v) \quad \text{and} \quad \frac{\partial z}{\partial y} = -2yf_u(u, v) + 2xf_v(u, v)$$

$$\begin{aligned}\frac{\partial^2 z}{\partial x^2} &= 2f_u(u, v) + 2x(2xf_{uu}(u, v) + 2yf_{uv}(u, v)) \\ &\quad + 2y(2xf_{vu}(u, v) + 2yf_{vv}(u, v)) \\ &= 2f_u(u, v) + 4x^2 f_{uu}(u, v) + 8xyf_{uv}(u, v) + 4y^2 f_{vv}(u, v),\end{aligned}$$

Similarly,

$$\begin{aligned}\frac{\partial^2 z}{\partial y^2} &= -2f_u(u, v) - 2y(-2yf_{uu}(u, v) + 2xf_{uv}(u, v)) \\ &\quad + 2x(-2yf_{vu}(u, v) + 2xf_{vv}(u, v)) \\ &= -2f_u(u, v) + 4y^2 f_{uu}(u, v) - 8xyf_{uv}(u, v) + 4x^2 f_{vv}(u, v).\end{aligned}$$

Therefore,

$$\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = 4(x^2 + y^2)(f_{uu}(u, v) + f_{vv}(u, v)) = 0$$

because f is harmonic. Thus, $z = f(x^2 - y^2, 2xy)$ is a harmonic function of x and y .

Example 17. If $z = f(x, y)$ has continuous partial derivatives of second order, and if $x = r \cos(\theta)$ and $y = r \sin(\theta)$, then find $\frac{\partial^2 z}{\partial r^2}$ and $\frac{\partial^2 z}{\partial \theta^2}$

Solution. Note that, $x = r \cos(\theta)$ and $y = r \sin(\theta)$ imply that

$$\frac{\partial x}{\partial r} = \cos \theta, \quad \frac{\partial x}{\partial \theta} = -r \sin \theta, \quad \frac{\partial y}{\partial r} = \sin \theta, \quad \frac{\partial y}{\partial \theta} = r \cos \theta.$$

Thus,

$$\frac{\partial z}{\partial r} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial r} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial r} = \cos \theta \frac{\partial z}{\partial x} + \sin \theta \frac{\partial z}{\partial y}.$$

$$\begin{aligned} \frac{\partial^2 z}{\partial r^2} &= \cos \theta \frac{\partial}{\partial r} \frac{\partial z}{\partial x} + \sin \theta \frac{\partial}{\partial r} \frac{\partial z}{\partial y} \\ &= \cos \theta \left(\cos \theta \frac{\partial^2 z}{\partial x^2} + \sin \theta \frac{\partial^2 z}{\partial x \partial y} \right) + \sin \theta \left(\cos \theta \frac{\partial^2 z}{\partial x \partial y} + \sin \theta \frac{\partial^2 z}{\partial y^2} \right) \\ &= \cos^2 \theta \frac{\partial^2 z}{\partial x^2} + 2 \cos \theta \sin \theta \frac{\partial^2 z}{\partial x \partial y} + \sin^2 \theta \frac{\partial^2 z}{\partial y^2}. \end{aligned}$$

Example 17. Cont'd Similarly,

$$\frac{\partial z}{\partial \theta} = -r \sin \theta \frac{\partial z}{\partial x} + r \cos \theta \frac{\partial z}{\partial y}.$$

and

$$\begin{aligned}\frac{\partial^2 z}{\partial \theta^2} &= -r \left(\cos \theta \frac{\partial z}{\partial x} + \sin \theta \frac{\partial}{\partial \theta} \frac{\partial z}{\partial x} \right) + r \left(-\sin \theta \frac{\partial z}{\partial y} + \cos \theta \frac{\partial}{\partial \theta} \frac{\partial z}{\partial y} \right) \\&= -r \frac{\partial z}{\partial r} - r \sin \theta \left(-r \sin \theta \frac{\partial^2 z}{\partial x^2} + r \cos \theta \frac{\partial z}{\partial y \partial x} \right) \\&\quad + r \cos \theta \left(-r \sin \theta \frac{\partial z}{\partial x \partial y} + r \cos \theta \frac{\partial^2 z}{\partial y^2} \right) \\&= -r \frac{\partial z}{\partial r} + r^2 \left(\sin^2 \theta \frac{\partial^2 z}{\partial x^2} - 2 \sin \theta \cos \theta \frac{\partial z}{\partial x \partial y} + \cos^2 \theta \frac{\partial^2 z}{\partial y^2} \right).\end{aligned}$$